The empirical recurrence for OEIS A189146, proved by transfer matrix

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6 September 2026

Abstract

OEIS A189146 counts the permutations of the cells of an $n \times 3$ array in which every cell moves by one of a fixed short list of offsets. The entry carries an empirical recurrence of order 58 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The count is the permanent of the incidence matrix of the offset graph; read the cells in row-major order and let each choose its image, and the only thing that has to be carried is which images inside a window of 15 consecutive positions are already taken. That makes a a walk count on $S = 1254$ states, hence C-finite, and whether the published recurrence is one of its recurrences is then a finite exact computation.

2020 Mathematics Subject Classification. 05A05, 05A15, 15A15.

1 The conjecture

OEIS A189146 is "Number of n X 3 array permutations with each element making zero or one knight moves.". It has offset 1 and begins

$$1, 4, 49, 569, 4372, 42689, 412189, 3988132, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 9*a(n-1) +48*a(n-2) -371*a(n-3) +173*a(n-4) -4636*a(n-5) -21700*a(n-6) +321034*a(n-7) -261016*a(n-8) -939316*a(n-9) +8804712*a(n-10) -89239632*a(n-11) +89709648*a(n-12) +830637056*a(n-13) -2499914752*a(n-14) +6390654336*a(n-15) -987233536*a(n-16) -98495866880*a(n-17) +227291122176*a(n-18) -22499119616*a(n-19) -574642074624*a(n-20) +4090433287168*a(n-21) -8710999644160*a(n-22) -9262792159232*a(n-23) +33961426997248*a(n-24) -59705619185664*a(n-25) +164268932415488*a(n-26) +206996041138176*a(n-27) -959952477028352*a(n-28) +148261211865088*a(n-29) +225201818959872*a(n-30) -528765464084480*a(n-31) +2794884482203648*a(n-32) +3264721555816448*a(n-33) +9397359024799744*a(n-34) -32363504358916096*a(n-35) -14560459025809408*a(n-36) +41205664512475136*a(n-37) -65070379105779712*a(n-38) +117763114509795328*a(n-39) -26803976066301952*a(n-40) +132124530591137792*a(n-41) +236469835482005504*a(n-42) -1308758689225637888*a(n-43) +382298681149227008*a(n-44) +1688786500906385408*a(n-45) -1092767223451222016*a(n-46) -1774519408253730816*a(n-47) +3781671287689052160*a(n-48) -211699968811991040*a(n-49) -6414401302863806464*a(n-50) +3737846953229156352*a(n-51) +2510967898491584512*a(n-52) +740560663725735936*a(n-53) -4415779434636771328*a(n-54) +2508504992445366272*a(n-55) +54043195528445952*a(n-56) -1152921504606846976*a(n-57) +576460752303423488*a(n-58)

As of the "Last modified" line on the live entry (May 06 2014 02:32:05, revision 13) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Write the cells of the array as pairs (i, j) with $0 \le i < n$ and $0 \le j < 3$. An *array permutation* of the kind the entry counts is a bijection π from the cells to the cells such that for every cell,

$$\pi(i,j) - (i,j) \in \Delta, \qquad \Delta = \{\, (-2,-1),\ (-2,1),\ (-1,-2),\ (-1,2),\ (0,0),\ (1,-2),\ (1,2),\ (2,-1),\ (2,1)\,\}.$$

The offsets are what the name's words say: the eight knight moves. A cell is allowed to stay where it is, and $(0, 0)$ is one of the offsets.

A move is only allowed when its target is a cell of the array, so cells near an edge have fewer choices. The reading is pinned against the entry's own published terms rather than assumed: taking "diagonally, horizontally or vertically" to mean all eight king moves gives 24 where A189305 publishes 14.

Equivalently, $a(n)$ is the permanent of the $n3 \times n3$ matrix whose (c, c') entry is 1 exactly when $c' - c \in \Delta$.

3 A window of 15 positions

Number the cells in row-major order, $k = i \cdot 3 + j$. An offset (δ_1, δ_2) moves a cell by $\delta_1 3 + \delta_2$ places in that order, so every image lies between $lo = -7$ and $hi = 7$ places from its source. Process the cells in order and let each one choose its image. Once the cell at position k has been dealt with, the position $k + -7$ can never be chosen again by anybody, so it must already be taken; and no position beyond $k + 7$ can have been taken yet. What has to be remembered is therefore exactly which of the 15 positions $k + -7, \ldots, k + 7$ are taken — a subset of a set of size 15, and nothing else.

That is the state. One step of the walk is one row, that is 3 cell choices, and the condition "the position leaving the window is taken" is what makes the walk count permutations rather than partial injections: every position is taken exactly once, once at the moment it leaves the window.

The two boundaries are the same condition. Images above the first row do not exist, so the walk starts with exactly the window positions lying above the array marked as taken. Images below the last row do not exist either, so the walk must end with exactly those positions marked and nothing beyond. In the window's own coordinates those two masks are equal, so the start state is also the only accepting state and

$$a(n) = \iota^\top M^j \tau, \qquad S = 1254,$$

for the transfer matrix M on those S states. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 1. *Let $q(t) = t^{58} - \sum_i c_i t^{58-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+58} - \sum_i c_i M^{j+58-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) = 9\,a(n-1)+48\,a(n-2)-371\,a(n-3)+173\,a(n-4)-4636\,a(n-5)-21700\,a(n-6)+321034\,a(n-7)-261016$$

for every $n > 57$.

Proof. The vector $w = q(M)\tau$ was formed by 58 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^{j} w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 57 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: the cells of the array were matched to their images one at a time, over the whole grid, with no window, no mask over a sliding frame and no transfer matrix, and the published terms came back. That computation shares nothing with the construction of Section 3 beyond the offset list.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A189146.

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[3] H. Minc, *Permanents*, Encyclopedia of Mathematics and its Applications 6, Addison–Wesley, 1978.